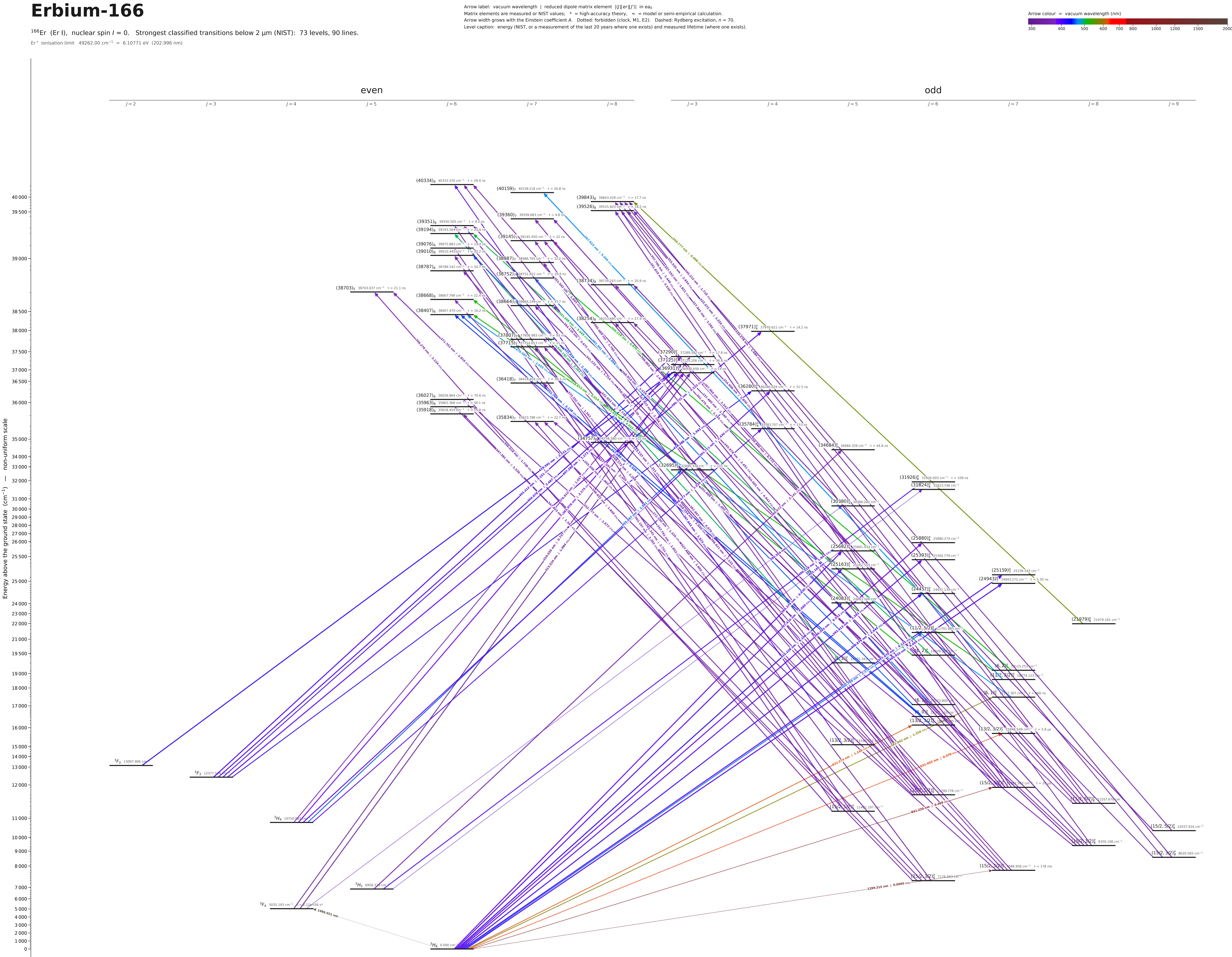




Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	U erI ² (ea)	A (s ⁻¹)	f/2π up. (MHz)	branch (%)	Use
³ H ₆ - (24943) ³ F ₂	400.9097	400.7964	747.7805	9.445	1.870e+08	29.75	100	broad cooling line: Zeeman slower, transverse cooling, im...
³ H ₆ - (6, 1) ³ F ₃	582.8421	582.6805	514.3631	1.310	1.170e+06	0.1851	100	narrow-line MOT
³ H ₆ - (13/2, 3/2) ³ F ₄	631.0522	630.8778	475.0676	0.579	1.800e+05	0.02842	100	narrow line proposed for cooling / spin manipulation
³ H ₆ - (15/2, 5/2) ³ F ₄	841.2196	840.9884	356.3784	0.469	5.000e+04	0.007958	100	8 kHz narrow-line cooling / narrow-line MOT
³ H ₆ - (15/2, 3/2) ³ F ₄	1299.2149	1298.8596	230.7489	0.010	5.600e+00	8.941e-07	99.7	ultranarrow innershell
³ H ₆ - ³ F ₄	1986.0212	1985.4791	150.9513	-	4.700e-05	7.493e-12	99.8	proposed E2 optical clock transition within the 4f12 6s2...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
f/2π: natural linewidth of the upper level, 1/(2πτ).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
³ H ₆ - (24943) ³ F ₂	400.910	170-166	-1681(14)
³ H ₆ - (24943) ³ F ₂	400.910	168-166	-840(14)
³ H ₆ - (24943) ³ F ₂	400.910	167-166	-297(6)
³ H ₆ - (24943) ³ F ₂	400.910	164-166	849(17)
³ H ₆ - (24943) ³ F ₂	400.910	170-168	-866.5(15)
³ H ₆ - (24943) ³ F ₂	400.910	166-164	-859.3(16)
³ H ₆ - (24943) ³ F ₂	400.910	164-162	-1028.7(28)
³ H ₆ - (6, 1) ³ F ₃	582.842	170-166	-1966(14)
³ H ₆ - (6, 1) ³ F ₃	582.842	168-166	-975(15)
³ H ₆ - (6, 1) ³ F ₃	582.842	167-166	-337(11)
³ H ₆ - (15/2, 3/2) ³ F ₄	1299.215	164-168	-2732.290(3)
³ H ₆ - (15/2, 3/2) ³ F ₄	1299.215	166-168	-1371.710(3)
³ H ₆ - (15/2, 3/2) ³ F ₄	1299.215	170-168	1414.920(5)

Shift = νfirst (isotope) - νsecond (isotope).